Theory of Value

An Axiomatic Analysis of Economic Equilibrium
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Chapter 1: Mathematics

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Limits in $\mathbb{R}^N$

The set $\mathbb{R}^N$ is the product of N sets equal to $\mathbb{R}$, i.e, $\mathbb{R} \times \cdots \times \mathbb{R} \times \cdots \times \mathbb{R}$. According to the general definitions of 1.2.11, an element, or point, x of $\mathbb{R}^N$ is an N-tuple of real numbers $x = (x_n) = (x_1, \cdots, x_n, \cdots, x_N)$. The nth number in the N-tuple is the nth coordinate of x.

Limits in $\mathbb{R}^N$

The set $\mathbb{R}^2$ may be visualized as follows. Draw in a plane a horizontal and a vertical straight line. The first (resp. second) set $\mathbb{R}$ in $\mathbb{R} \times \mathbb{R}$ will be represented by the horizontal (resp. vertical) line by choosing 0 at the intersection point, and 1 arbitrarily to the right of 0 (resp. above 0). Then an element (x_1, x_2) of $\mathbb{R}^2$ is represented by the intersection point of the vertical through x_1 and the horizontal through x_2.

Limits in $\mathbb{R}^N$

To visualize $\mathbb{R}^3$, consider, in ordinary space, the west-east, the south-north, and the vertical straight line through a point 0. They will represent respectively the first, the second, and the third set in $\mathbb{R} \times \mathbb{R} \times \mathbb{R}$. An element (x_1, x_2, x_3) of $\mathbb{R}^3$ is represented by the intersection point of the vertical south-north plane through x_1, the vertical west-east plane through x_2, and the horizontal plane through x_3.

Limits in $\mathbb{R}^N$

$\mathbb{R}^N$ is called the *N*-dimensional Euclidean space. Let $\mathscr{N}$ be a subset of the set of the first N positive integers $\{1, \cdots, n, \cdots, N\}$. The set $\left\{x \in \mathbb{R}^N \mid x_n = 0, n \notin \mathscr{N}\right\}$ is called a coordinate subspace of $\mathbb{R}^N$. Its dimension is the number of elements of $\mathscr{N}$. The point $\mathbf{0}$, all of whose coordinates are equal to 0, is the origin of $\mathbb{R}^N$.

Sets in $\mathbb{R}^N$

N-dimensional Euclidean space, or N-dimensional continuum, is the set $\mathbb{R}^N$ of all number tuple of N real numbers.

$$\mathbb{R}^N = \{x = (x_1, \cdots, x_N) : x_n \in \mathbb{R} \text{ for } n = 1, \cdots N\}.$$

In general, for any two points $x^k = (x_1^k, \cdots, x_N^k)$ in $\mathbb{R}^N$ for $k = 1, 2$, we define the distance (Euclidean l^p metric) between the two points for $1 \leqslant p < \infty$

$$\rho_p(x^1, x^2) = \left\{ \sum_{n=1}^{N} \left| x_n^1 - x_n^2 \right|^p \right\}^{\frac{1}{p}}.$$

The useful metrics on $\mathbb{R}^N$ include the taxicab (or l^1) metric for $p = 1$

$$\rho_1(x^1, x^2) = \sum_{n=1}^{N} \left| x_n^1 - x_n^2 \right|,$$

the distance (Euclidean l^2 metric) between the two points for $p = 2$

$$\rho_2(x^1, x^2) = \left\{ \sum_{n=1}^{N} \left(x_n^1 - x_n^2 \right)^2 \right\}^{\frac{1}{2}},$$

and the max (or l^∞) metric for $p = \infty$

$$\rho_\infty(x^1, x^2) = \max \left\{ \left| x_n^1 - x_n^2 \right| : 1 \leqslant n \leqslant N \right\}.$$

Sets in $\mathbb{R}^N$

$$\lim_{p\to\infty} \rho_p(x^1,x^2) = \lim_{p\to\infty} \left\{ \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p \right\}^{\frac{1}{p}}$$

$$= \lim_{p\to\infty} e^{\frac{1}{p}\ln \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p} = e^{\lim_{p\to\infty} \frac{1}{p}\ln \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p}$$

$$= e^{\lim_{p\to\infty} \frac{\sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p \ln\left|x_n^1 - x_n^2\right|}{\sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|^p}} = e^{\sum_{n=1}^{N} \lim_{p\to\infty} \frac{\left|x_n^1 - x_n^2\right|^p}{\sum_{n'=1}^{N} \left|x_{n'}^1 - x_{n'}^2\right|^p} \ln\left|x_n^1 - x_n^2\right|}$$

$$= e^{\max_{1\leqslant n\leqslant N} \ln\left|x_n^1 - x_n^2\right|} = \max_{1\leqslant n\leqslant N} \left|x_n^1 - x_n^2\right|.$$

Sets in $\mathbb{R}^N$

Then we have the following three propositions.

(1) Positivity: $\rho(x^1, x^2) \geqslant 0$. $\rho(x^1, x^2) = 0$ if and only if $x^1 = x^2$ (i.e. $x_n^1 = x_n^2$ for $n = 1, \cdots N$).

(2) Symmetry: $\rho(x^1, x^2) = \rho(x^2, x^1)$.

(3) Triangle Inequality: $\rho(x^1, x^2) \leqslant \rho(x^1, x^0) + \rho(x^0, x^2)$.

From the triangle inequality, $\rho(x^1, x^2)$ is a continuous function on (x^1, x^2): If $\lim_{n\to\infty} \rho(x_n^1, x^1) = 0$ and $\lim_{n\to\infty} \rho(x_n^2, x^2) = 0$, then $\lim_{n\to\infty} \rho(x_n^1, x_n^2) = \rho(x^1, x^2)$. In particular,

$$\lim_{\rho(x_n^2, x^2)\to 0} \rho(x^1, x_n^2) = \rho(x^1, x^2) \text{ and } \lim_{\rho(x_n^1, x^1)\to 0} \rho(x_n^1, x^2) = \rho(x^1, x^2).$$

Sets in $\mathbb{R}^N$

The set of all points in $\mathbb{R}^N$ whose distance to some given point x_0 is strictly less than δ is call neighborhood with center x_0 and radius δ (without boundary), written as $N(x_0, \delta)$.

$$N(x_0, \delta) = \{x \in \mathbb{R}^N : \rho(x, x_0) < \delta\}.$$

When it must not mention point x_0 or radius δ, we just say a neighborhood. If the boundary is required, we define the neighborhood with boundary. The set of all points in $\mathbb{R}^N$ whose distance to some given point x_0 is less than δ is call neighborhood with center x_0 and radius δ (with boundary), written as $B(x_0, \delta)$.

$$B(x_0, \delta) = \{x \in \mathbb{R}^N : \rho(x, x_0) \leqslant \delta\}.$$

Sometimes we do not need the center x_0 in neighborhood with center x_0 and radius δ, $B(x_0, \delta)$. The set of all points in $\mathbb{R}^N$ whose distance to some given point x_0 is strictly less than δ is call neighborhood with center x_0 and radius δ without center x_0, written as $D(x_0, \delta)$.

$$\begin{aligned} D(x_0, \delta) &= \{x \in \mathbb{R}^N : 0 < \rho(x, x_0) < \delta\} \\ &= \{x \in \mathbb{R}^N : \rho(x, x_0) < \delta, x \neq x_0\} \\ &= N(x_0, \delta) \setminus \{x_0\} \end{aligned}$$

which is neighborhood with center x_0 and radius δ deleting center x_0.

Sets in $\mathbb{R}^N$

For any point $x = (x_1, \cdots, x_N)$ in $\mathbb{R}^N$, define its norm or metric $\|x\|$ to present $\rho(x, 0)$, the distance (Euclidean l^p metric) for $1 \leqslant p < \infty$

$$\|x\|_p = \rho_p(x, 0) = \left\{ \sum_{n=1}^{N} |x_n|^p \right\}^{\frac{1}{p}}.$$

The useful metrics on $\mathbb{R}^N$ include the taxicab (or l^1) metric for $p = 1$

$$\|x\|_1 = \rho_1(x, 0) = \sum_{n=1}^{N} |x_n|,$$

Euclidean l^2 metric for $p = 2$

$$\|x\|_2 = \rho_2(x, 0) = \left\{ \sum_{n=1}^{N} [x_n]^2 \right\}^{\frac{1}{2}},$$

and the max (or l^∞) metric for $p = \infty$

$$\|x\|_\infty = \rho_\infty(x, 0) = \max\{|x_n| : 1 \leqslant n \leqslant N\},$$

i.e., the greatest of the absolute values of its coordinates

Sets in $\mathbb{R}^N$

$$\|x\|_{p'} \leqslant \|x\|_p \leqslant N^{\frac{1}{p}-\frac{1}{p'}} \|x\|_{p'}, \qquad 1 \leqslant p \leqslant p' \leqslant \infty$$

$$\|x\|_{p'} \leqslant \|x\|_1 \leqslant N^{1-\frac{1}{p'}} \|x\|_{p'}, \qquad 1 = p < p' \leqslant \infty$$

$$\|x\|_{p'} \leqslant \|x\|_p \leqslant N^{\frac{1}{p}-\frac{1}{p'}} \|x\|_{p'}, \qquad 1 < p < p' < \infty$$

$$\|x\|_{\infty} \leqslant \|x\|_p \leqslant N^{\frac{1}{p}} \|x\|_{\infty}, \qquad 1 \leqslant p < p' = \infty$$

Then we have the following three propositions.

(1) Positivity: $\|x\| \geqslant 0$. $\|x\| = 0$ if and only if $x = 0$ (i.e. $x_n = 0$ for $n = 1, \cdots N$).

(2) Homogeneity (of Degree 1): $\|\alpha x\| = |\alpha| \|x\|$.

(3) Triangle Inequality: $\|x^1 + x^2\| \leqslant \|x^1\| + \|x^2\|$.

Sets in $\mathbb{R}^N$

E is a point set of $\mathbb{R}^N$, if there exists a constant M such that, for any $x = (x_1, \cdots, x_N) \in E$, $\|x\| \leqslant M$, then we say that E is a bounded set. For example, for $1 \leqslant p \leqslant \infty$, $\|x\|_p = \left\{\sum_{n=1}^{N} |x_n|^p\right\}^{\frac{1}{p}} \leqslant M$ for any $x = (x_1, \cdots, x_N) \in E$, then E is a bounded set. If $p = \infty$, E is a bounded set if and only if, for $x \in E$, there exists a constant M such that $\|x\|_\infty = \max\{|x_1|, \cdots, |x_N|\} \leqslant M$, that is, $|x_n| \leqslant M$ for $n = 1, \cdots, N$. Let M be a non-negative real number; the set $\mathbb{M} = \left\{x \in \mathbb{R}^N \mid \|x\| \leqslant M\right\}$ is a closed cube of $\mathbb{R}^N$ with center 0, edge $2M$.

Limits in $\mathbb{R}^N$

Consider a sequence (x^q) of points of $\mathbb{R}^N$. One says that (x^q) converges (or tends) to a point x^0 of $\mathbb{R}^N$ if, for any $\varepsilon > 0$, there exists a large integer number $Q > 0$ such that, when $q > Q$, $\|x^q - x^0\| < \varepsilon$. One writes $x^q \to x^0$.

For the norm or metric $\|x\|_p = \left\{\sum_{n=1}^N |x_n|^p\right\}^{\frac{1}{p}}$ for $1 \leqslant p \leqslant \infty$, the sequence (x^q) converges (or tends) to a point x^0 of $\mathbb{R}^N$ if, for any $\varepsilon > 0$, there exists a large integer number $Q > 0$ such that, when $q > Q$, $\|x^q - x^0\|_p < \varepsilon$ or equivalently $\|x^q - x^0\|_\infty = \max\left\{\left|x_1^q - x_1^0\right|, \cdots, \left|x_N^q - x_N^0\right|\right\} < \varepsilon$, that is, $\|x_n^q - x_n^0\|_\infty < \varepsilon$ for $n = 1, \cdots, N$, i.e., for all coordinates $(n = 1, \cdots, N)$, one has $x_n^q \to x_n^0$.

Limits in $\mathbb{R}^N$

As in the case of numbers (1.5.o), a sequence which tends to a point is called convergent; the unique point to which a convergent sequence tends is called its limit.

Accumulation Points, Interior Points, and Boundary Points

Let E be a set of N-dimensional Euclidean space $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$, we now explore the relationship between P_0 and E: there exist three possible settings.

(1) There are not any points of E around P_0. There exists a neighborhood $N(P_0, \delta)$ with center P_0 such that there are not any points of $N(P_0, \delta)$ in E: $N(P_0, \delta) \cap E = \varnothing$.

(2) All points around P_0 are in E. There exists a neighborhood $N(P_0, \delta)$ with center P_0 such that $N(P_0, \delta) \subseteq E$, then P_0 is one of interior points of E.

(3) Some points around P_0 are in E and some are not in. There exists a neighborhood $N(P_0, \delta)$ with center P_0 such that, for some $P \in N(P_0, \delta)$, $P \in E$, and for some $P' \in N(P_0, \delta)$, $P' \notin E$, then P_0 is one of boundary points of E.

Accumulation Points, Interior Points, and Boundary Points

Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. If there exists a neighborhood $N(P_0, \delta)$ with center P_0 such that $N(P_0, \delta) \subseteq E$, then P_0 is one of interior points of E.

Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. If in any neighborhood $N(P_0, \delta)$ with center P_0 there exist infinite points in $N(P_0, \delta)$ belong to E, then P_0 is one of accumulation points (cluster points or limit points) of E.

Any interior point of E is an accumulation point of E. However, an accumulation point of E must not be a interior point of E. In fact, it is possible one of boundary points of E. Any interior point of E belongs to E. However, an accumulation point of E may belong to E, or may not belong to E.

Let E be a point set of $\mathbb{R}^N$. The set of all accumulation points of E is called the derived set, written as E'. The set of all interior points of E is called the interior, written as E°. $E \cup E'$ is call closure of E, written as $\overline{E} = E \cup E'$.

Accumulation Points, Interior Points, and Boundary Points

Theorem 1. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E'$ if and only if P_0 is one of limit points of E, or equivalently, there exists a sequence of different points $P_n \in E$ such that $\lim_{n\to\infty} \rho(P_n, P_0) = 0$.

Proof. [Sufficiency] For any neighborhood $N(P_0, \delta)$ of center P_0 with $\delta > 0$, then, if N is large enough, $\rho(P_n, P_0) < \delta$ for $n \geqslant N$, then $P_n \in N(P_0, \delta)$, and there exist infinite such P_n.

[Necessity] Define $\delta_n = \dfrac{1}{n}$, then $\delta_n > 0$. $P_0 \in E'$, and hence infinite points in $N(P_0, \delta_1)$ belong to E. We can take P_1 arbitrarily. $P_0 \in E'$, and hence infinite points in $N(P_0, \delta_2)$ belong to E, thus $N(P_0, \delta_2) \setminus \{P_1\} \neq 0$. We can take $P_2 \in N(P_0, \delta_2) \setminus \{P_1\}$ arbitrarily, both P_1 and P_2 are different. Generally, we assume that we take different $P_1, P_2, \cdots, P_N$ with $P_n \in N(P_0, \delta_n)$ and $P_n \in E$ for $n = 1, \cdots, N$. Then infinite points in $N(P_0, \delta_{N+1})$ belong to E, thus $N(P_0, \delta_{N+1}) \setminus \{P_1, P_2, \cdots, P_N\} \neq 0$. We can take $P_{N+1} \in N(P_0, \delta_{N+1}) \setminus \{P_1, P_2, \cdots, P_N\}$ arbitrarily, P_{N+1} is different from $P_1, P_2, \cdots, P_N$. Therefore, we obtain a sequence of different points $P_n \in E$ for $n = 1, 2, \cdots$ such that $P_n \in N(P_0, \delta_n)$ and $P_n \in E$ for $n = 1, 2, \cdots$. Note $\lim_{n\to\infty} \delta_n = 0$, then $\lim_{n\to\infty} \rho(P_n, P_0) = 0$.

Q.E.D.

Accumulation Points, Interior Points, and Boundary Points

Theorem 2. If $A_1 \subseteq A_2$, then $A_1' \subseteq A_2'$ and $\overline{A}_1 \subseteq \overline{A}_2$.

Theorem 3. $(A_1 \cup A_2)' = A_1' \cup A_2'$.

Proof. Note $A_1 \subseteq A_1 \cup A_2$ and $A_2 \subseteq A_1 \cup A_2$, then $A_1' \subseteq (A_1 \cup A_2)'$ and $A_2' \subseteq (A_1 \cup A_2)'$ from Theorem 2, and hence $A_1' \cup A_2' \subseteq (A_1 \cup A_2)'$.

If $P \in (A_1 \cup A_2)'$, then there exists a sequence of different points $P_n \in A_1 \cup A_2$ such that $\lim\limits_{n\to\infty} \rho(P_n, P) = 0$. If $P \in A_1'$, then $P \in A_1' \cup A_2'$. If $P \notin A_1'$, then at most finite P_n for $n = 1, 2, \cdots$ belong to A_1, the remaining infinite points belong to A_2, then $P \in A_2'$ from Theorem 1. Thus $P \in A_1' \cup A_2'$. Therefore $(A_1 \cup A_2)' \subseteq A_1' \cup A_2'$. Q.E.D.

Corollary 1. $\left(\bigcup\limits_{n=1}^{N} A_n\right)' = \bigcup\limits_{n=1}^{N} A_n'$.

Accumulation Points, Interior Points, and Boundary Points

Theorem 4. If $A_1 \subseteq A_2$, then $A_1^\circ \subseteq A_2^\circ$.

Theorem 5. $(A_1 \cap A_2)^\circ = A_1^\circ \cap A_2^\circ$.

Proof. Note $A_1 \cap A_2 \subseteq A_1$ and $A_1 \cap A_2 \subseteq A_2$, then $(A_1 \cap A_2)^\circ \subseteq A_1^\circ$ and $(A_1 \cap A_2)^\circ \subseteq A_2^\circ$ from Theorem 2, and hence $(A_1 \cap A_2)^\circ \subseteq A_1^\circ \cap A_2^\circ$.

If $P \in A_1^\circ \cap A_2^\circ$, then $P \in A_k^\circ$ for $k = 1, 2$, and there exists a a neighborhood $N(P, \delta)$ of center P such that $N(P, \delta) \subseteq A_k$ for $k = 1, 2$, thus $N(P, \delta) \subseteq A_1 \cap A_2$. Therefore $P \in (A_1 \cap A_2)^\circ$, and hence $A_1^\circ \cap A_2^\circ \subseteq (A_1 \cap A_2)^\circ$. Q.E.D.

Corollary 2. $\left(\bigcap_{n=1}^{N} A_n \right)^\circ = \bigcap_{n=1}^{N} A_n^\circ$.

Bolzano-Weierstrass Theorem

Theorem 6 (Bolzano-Weierstrass Theorem). Let E be a bounded infinite set in N-dimensional Euclidean space $\mathbb{R}^N$. Then there exists at least an accumulation point P of E (The point P may not belong to E), i.e. $E' \neq \varnothing$.

Bolzano-Weierstrass Theorem

Proof. Note that E is a bounded infinite set, there exists a constant M such that, for any $x = (x_1, \cdots, x_N) \in E$, $|x_n| \leqslant M$ for $n = 1, \cdots, N$, and hence E includes in N-dimensional cube

$$R_0 = E[(x_1, \cdots, x_N); |x_n| \leqslant M \text{ for } n = 1, \cdots, N].$$

We use axises to divide R_0 into 2^N small closed cubes, then at least one small closed cube includes infinite points in E (otherwise E is a finite set!), we call such a small closed cube R_1 with edge M. Generally, we assume that there exists a sequence of closed cubes whose edges are parallel to axises.

$$R_0 \supseteq R_1 \supseteq \cdots \supseteq R_N, \quad R_N \cap E \text{ is an infinite set,}$$
$$\text{the edge of } R_n \text{ is } \frac{M}{2^{n-1}} \text{ for } n = 1, 2, \cdots, N$$

Bolzano-Weierstrass Theorem

We use lines parallel to axises to divide R_N into 2^N small closed cubes, then at least one small closed cube includes infinite points in E (otherwise E is a finite set!), we call such a small closed cube R_{N+1} with edge $\frac{M}{2^N}$. Thus we obtain a sequence of decreasing closed cubes whose edges are parallel to axises.

$$R_0 \supseteq R_1 \supseteq \cdots \supseteq R_n \supseteq \cdots, \quad R_n \cap E \text{ is an infinite set,}$$
$$\text{and the edge of } R_n \text{ is } \frac{M}{2^{n-1}} \text{ for } n = 1, 2, \cdots$$

Note $\lim\limits_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then, from Cantor Nested Intervals Theorem, there exists a unique point $P \in \bigcap\limits_{n=1}^{\infty} R_n$. We now check that this P is one of accumulation points of E.

Define $N(P, \delta)$ as any neighborhood of center P. Note $\lim\limits_{n\to\infty} \frac{M}{2^{n-1}} = 0$, then for enough large natural number n_0, $\frac{M}{2^{n_0}} < \frac{\delta}{n_0{}^{\frac{1}{2}}}$, $n_0{}^{\frac{1}{2}} \frac{M}{2^{n_0}} < \delta$,. The diagonal length of R_{n_0+1} is strictly less than δ, then $R_{n_0+1} \subseteq N(P, \delta)$, and hence $E \cap R_{n_0+1} \subseteq E \cap N(P, \delta)$. There exist infinite points of $N(P, \delta)$ in E.
Q.E.D.

Bolzano-Weierstrass Theorem

In mathematics, a point P is called an isolated point of a set E if P is an element of E but there exists a neighborhood of P which does not contain any other points of E.

Point P is an isolated point of set E if $P \in E$ and $P \notin D(P, \delta)$ for $\delta > 0$.

The boundary point which is not accumulation point is called to be an isolated point. The isolated point belongs to E. A set E is isolated if every point is isolated in E.

Each isolated set is a countable (finite or denumerable) set. In fact, if E is an isolated set, then, for any $P \in E$, there exists $\delta_P > 0$ such that $N(P, 2\delta_P) \cap E = \{P\}$. We pick a rational point R_P in $N(P, \delta_P)$. If $P_1 \neq P_2$, then $R_{P_1} \neq R_{P_2}$. Otherwise, $R_{P_1} = R_{P_2}$, then

$$\begin{aligned} \rho(P_1, P_2) &\leqslant \rho(P_1, R_{P_1}) + \rho(R_{P_1}, P_2) = \rho(P_1, R_{P_1}) + \rho(P_2, R_{P_2}) \\ &< \delta_{P_1} + \delta_{P_2} \leqslant 2 \max\{\delta_{P_1}, \delta_{P_2}\}. \end{aligned}$$

Bolzano-Weierstrass Theorem

(1) If $\max\{\delta_{P_1}, \delta_{P_2}\} = \delta_{P_1}$, then $\rho(P_1, P_2) < 2\delta_{P_1}$, then $P_2 \in N(P_1, 2\delta_{P_1})$ and $N(P_1, 2\delta_{P_1}) \cap E = \{P_1, P_2\}$.

(2) If $\max\{\delta_{P_1}, \delta_{P_2}\} = \delta_{P_2}$, then $\rho(P_1, P_2) < 2\delta_{P_2}$, then $P_1 \in N(P_2, 2\delta_{P_2})$ and $N(P_2, 2\delta_{P_2}) \cap E = \{P_1, P_2\}$.

Therefore there exists an one - to -one correspondence from $P \in E$ to R_P. Note that the set of all rationals is denumerable, then E is a countable (finite or denumerable) set.

E is an isolated set if and only if $E \cap E' = \varnothing$.

E is a discrete set if $E' = \varnothing$. Any discrete set is an isolated set, but an isolated set may not be a discrete set. For example, the point set $\left\{\frac{1}{n}; n = 1, 2, \cdots\right\} \subseteq \mathbb{R}$ is an isolated set, but it is not a discrete set because 0 is an accumulation point.

Accumulation Points, Interior Points, and Boundary Points

Theorem 7. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E'$ if and only if, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P').

Proof. [Necessity] If $P_0 \in E'$, then, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P').

If $P_0 \in E'$, then for any $\delta > 0$ there exist infinite many points in $N(P_0, \delta) \cap E$. Take $P \in N(P_0, \delta)$, then $0 \leqslant d = \rho(P, P_0) < \delta$. Thus for any $P' \in N(P_0, \delta - d)$ we have
$\rho(P, P') \leqslant \rho(P, P_0) + \rho(P_0, P') < d + \delta - d = \delta$, and hence
$N(P_0, \delta - d) \subseteq N(P, \delta)$ and $N(P_0, \delta - d) \cap E \subseteq N(P, \delta) \cap E$.

Note that $N(P_0, \delta - d) \cap E$ has infinite many points, then there exists a point P', differing from P_0, in $N(P_0, \delta - d) \cap E \subseteq N(P, \delta) \cap E$ (there exist infinite such points P').

Accumulation Points, Interior Points, and Boundary Points

[Sufficiency] If, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point P', differing from P_0, in E (there exist infinite such points P'), then $P_0 \in E'$.

In any neighborhood $N(P, \delta)$ of point P_0, there exists a point P_1, differing from P_0, in E. Write $0 < \delta_1 = \rho(P_1, P_0) < \delta$. In δ_1 neighborhood $N(P_0, \delta_1)$ of center P_0, there exists a point P_2, differing from P_0, in E. Write $0 < \delta_2 = \rho(P_2, P_0) < \delta_1$, then $P_2 \in N(P_0, \delta_1)$. Generally, in δ_n neighborhood $N(P_0, \delta_n)$ of center P_0, there exists a point P_{n+1}, differing from P_0, in E. Write $0 < \delta_{n+1} = \rho(P_{n+1}, P_0) < \delta_n$, then $P_{n+1} \in N(P_0, \delta_n)$. From induction, there exist $\{\delta_n : n \in \mathbb{N}\}$ with $\delta_{n+1} < \delta_n$ for $n \in \mathbb{N}$ such that point P_{n+1}, differing from P_0, in E. Write $0 < \delta_{n+1} = \rho(P_{n+1}, P_0) < \delta_n$, then $P_{n+1} \in N(P_0, \delta_n)$. Therefore there exist infinitely many points in $N(P_0, \delta) \cap E$, and hence $P_0 \in E'$. Q.E.D.

Accumulation Points, Interior Points, and Boundary Points

Theorem 8. Let E be a set of $\mathbb{R}^N$ and P_0 be a given point in $\mathbb{R}^N$. $P_0 \in E^\circ$ if and only if there exists a neighborhood $N(P, \delta)$ of point P_0 such that $N(P, \delta) \subseteq E$.

Proof. [Necessity] If $P_0 \in E^\circ$, then there exists a neighborhood $N(P, \delta)$ of point P_0 such that $N(P, \delta) \subseteq E$.
If $P_0 \in E^\circ$, then there exists $\delta > 0$ such that $N(P_0, \delta) \subseteq E$. Take $P \in N(P_0, \delta)$, then $0 \leqslant d = \rho(P, P_0) < \delta$. Thus for any $P' \in N(P, \delta - d)$ we have $\rho(P_0, P') \leqslant \rho(P, P_0) + \rho(P, P') < d + \delta - d = \delta$, and hence $P' \in N(P_0, \delta)$. Thus $N(P, \delta - d) \subseteq N(P_0, \delta)$ and $N(P, \delta - d) \subseteq E$.

[Sufficiency] If there exists a neighborhood $N(P, \delta)$ of point P_0 such that $N(P, \delta) \subseteq E$, then $P_0 \in E^\circ$.
There exists a neighborhood $N(P, \delta)$ of point P_0 such that $N(P, \delta) \subseteq E$, then $N(P_0, \delta - \rho(P_0, P)) \subseteq N(P, \delta) \subseteq E$, and hence $P_0 \in E^\circ$. Q.E.D.

Closed Set

Let $E \subseteq \mathbb{R}^N$. If $E' \subseteq E$, all accumulation points belong to E, then E is a closed set (here, the empty set is defined as a closed set). Note $\overline{E} = E \cup E'$, we say $\overline{E}$ is the closure E (E is a closed set if and only if $E = \overline{E}$).

Example 1. The closed rectangular in $\mathbb{R}^N$ is a closed set. $\mathbb{R}^N$ itself is a closed set. The closure of the rational number set $\mathbb{Q}$ is $\mathbb{R}$.

Example 2. If $f(x)$ is defined on $\mathbb{R}^N$, then the sufficient and necessary conditions for $f \in C\left(\mathbb{R}^N\right)$ are, for any $t \in \mathbb{R}$, the sets

$$E_1 = \left\{x \in \mathbb{R}^N \mid f(x) \geqslant t\right\} \qquad \text{and} \qquad E_2 = \left\{x \in \mathbb{R}^N \mid f(x) \leqslant t\right\}$$

are closed sets.

Example 3. The closure of the open ball $N(x_0, r)$ in $\mathbb{R}^N$ is the closed ball $B(x_0, r)$:

$$B(x_0, r) = \overline{N(x_0, r)} = \left\{x \in \mathbb{R}^N \mid |x - x_0| \leqslant r\right\}.$$

Closed Set

Example 2. If $f(x)$ is defined on $\mathbb{R}^N$, then the sufficient and necessary conditions for $f \in C\left(\mathbb{R}^N\right)$ are, for any $t \in \mathbb{R}$, the sets

$$E_1 = \left\{x \in \mathbb{R}^N \mid f(x) \geqslant t\right\} \qquad \text{and} \qquad E_2 = \left\{x \in \mathbb{R}^N \mid f(x) \leqslant t\right\}$$

are closed sets.

Proof [Necessity] For E_1, if $\{x_k; k \in \mathbb{N}\} \subseteq E_1$ and $\lim\limits_{k\to\infty} x_k = x_0$, from the continuity of $f(x)$ and $f\left(x_k\right) \geqslant t$, then $f\left(x_0\right) = \lim\limits_{k\to\infty} f\left(x_k\right) \geqslant t$, i.e. $x_0 \in E_1$. For E_2, if $\{x_k; k \in \mathbb{N}\} \subseteq E_2$ and $\lim\limits_{k\to\infty} x_k = x_0$, from the continuity of $f(x)$ and $f\left(x_k\right) \leqslant t$, then $f\left(x_0\right) = \lim\limits_{k\to\infty} f\left(x_k\right) \leqslant t$, i.e. $x_0 \in E_2$.

[Sufficiency] We adopt the reduction to absurdity. We assume that there exists $x_0 \in \mathbb{R}^N$ which is not a continuous point of $f(x)$, then there is $\varepsilon_0 > 0$ and point sequence $\{x_k; k \in \mathbb{N}\}$ such that $\lim\limits_{k\to\infty} x_k = x_0$. For every $k \in \mathbb{N}$,

$$f\left(x_k\right) \leqslant f\left(x_0\right) - \varepsilon_0 \qquad \text{or} \qquad f\left(x_k\right) \geqslant f\left(x_0\right) + \varepsilon_0.$$

If $f\left(x_k\right) \leqslant f\left(x_0\right) - \varepsilon_0$, then $x_k \in E_2[t = f\left(x_0\right) - \varepsilon_0]$, but $x_0 \notin E_2[t = f\left(x_0\right) - \varepsilon_0]$, which is a contradiction! If $f\left(x_k\right) \geqslant f\left(x_0\right) + \varepsilon_0$, then $x_k \in E_1[t = f\left(x_0\right) + \varepsilon_0]$, but $x_0 \notin E_1[t = f\left(x_0\right) + \varepsilon_0]$, which is a contradiction!

Closed Set

Example 3. The closure of the open ball $N(x_0, r)$ in $\mathbb{R}^N$ is the closed ball $B(x_0, r)$:

$$B(x_0, r) = \overline{N(x_0, r)} = \left\{ x \in \mathbb{R}^N \mid |x - x_0| \leqslant r \right\}.$$

Proof Define $F = \left\{ x \in \mathbb{R}^N \mid |x - x_0| \leqslant r \right\}$, then f is a closed set, so $\overline{N(x_0, r)} \subseteq F$. Conversely, if $x \in F$, define $x_k = \frac{1}{k} x_0 + \left(1 - \frac{1}{k}\right) x$, and

$$|x_0 - x_k| = \left(1 - \frac{1}{k}\right) |x - x_0| \leqslant \left(1 - \frac{1}{k}\right) r < r \text{ and } |x - x_k| = \frac{1}{k} |x - x_0| \leqslant \frac{1}{k} r < r.$$

This indicates that $\{x_k; k \in \mathbb{N}\} \subseteq N(x_0, r)$, and $\lim\limits_{k \to \infty} x_k = x$. Thus $x \in \overline{N(x_0, r)}$, that is, $F \subseteq \overline{N(x_0, r)}$.

Note: If $A \subseteq S$ and $\overline{A} = S$, then A is said to be dense in S, or A is a dense subset of S.

Closed Set

If with each subset X of a set S we associate a subset $\overline{X}$ of S such that

(1) $X \subseteq \overline{X}$,

(2) $X_1 \subseteq X_2 \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \overline{X}$,

(4) $\overline{\varnothing} = \varnothing$,

then we call the correspondence $X \to \overline{X}$ a closure operation.

Closed Set

Theorem 1. E' and $\overline{E}$ are closed sets.

Proof. If $P_0 \in (E')'$, then, in any neighborhood $N(P, \delta)$ of point P_0, there exists a point $P_1 \in E'$ with $P_1 \neq P_0$ from Theorem 7 in Subsection 2, and hence, in any neighborhood $N(P', \delta')$ of point P_1, there exists a point $P_2 \in E$ with $P_2 \neq P_1$ and $P_2 \neq P_0$ from Theorem 7 in Subsection 2, thus $P_0 \in E'$. Therefore E' is a closed set.

Note $\overline{E} = E \cup E'$, then $\overline{E}' = E' \cup (E')' \subseteq E' \cup E' = E' \subseteq \overline{E}$, and hence $\overline{E}$ is a closed set. Q.E.D.

Closed Set

Theorem 2. The union of finite closed sets is a closed set.

Proof. Define $F = \bigcup_{n=1}^{N} F_n$ where F_n is a closed set for $n = 1, \cdots, N$. Note that F_n is a closed set, $F_n' \subseteq F_n$, then

$$F' = \left(\bigcup_{n=1}^{N} F_n \right)' = \bigcup_{n=1}^{N} F_n' \subseteq \bigcup_{n=1}^{N} F_n = F.$$

Therefore F is a closed set. Q.E.D.

Closed Set

Theorem 2. The union of finite closed sets is a closed set.

Proof. Define $F = \bigcup_{n=1}^{N} F_n$ where F_n is a closed set for $n = 1, \cdots, N$. Note that F_n is a closed set, $\overline{F}_n = F_n \cup F_n'$, then

$$\begin{aligned}
\overline{F} &= \overline{\bigcup_{n=1}^{N} F_n} = \left(\bigcup_{n=1}^{N} F_n\right) \cup \left(\bigcup_{n=1}^{N} F_n\right)' \\
&= \left(\bigcup_{n=1}^{N} F_n\right) \cup \left(\bigcup_{n=1}^{N} F_n'\right) \\
&= \bigcup_{n=1}^{N} (F_n \cup F_n') = \bigcup_{n=1}^{N} \overline{F}_n = \bigcup_{n=1}^{N} F_n = F.
\end{aligned}$$

Therefore F is a closed set. Q.E.D.

Closed Set

Theorem 3. The intersection of multiple closed sets is a closed set.

Proof. Define $F = \bigcap_{n=1}^{\infty} F_n$ where F_n is a closed set for $n = 1, 2, \cdots$.

Note $F \subseteq F_n$, then $F' \subseteq F_n'$. Thus $F' \subseteq \bigcap_{n=1}^{\infty} F_n'$. Note that F_n is a closed set, $F_n' \subseteq F_n$, then

$$\left(\bigcap_{n=1}^{\infty} F_n\right)' = F' \subseteq \bigcap_{n=1}^{\infty} F_n' \subseteq \bigcap_{n=1}^{\infty} F_n = F.$$

Therefore F is a closed set. Q.E.D.

Closed Set

Theorem 3. The intersection of multiple closed sets is a closed set.

Proof. Define $F = \bigcap_{n=1}^{\infty} F_n$ where F_n is a closed set for $n = 1, 2, \cdots$.

Note $F \subseteq F_n$, then $\overline{F} \subseteq \overline{F}_n$. Thus $\overline{F} \subseteq \bigcap_{n=1}^{\infty} \overline{F}_n$. Note that F_n is a closed set, $\overline{F}_n = F_n$, then

$$\overline{\bigcap_{n=1}^{\infty} F_n} = \overline{F} \subseteq \bigcap_{n=1}^{\infty} \overline{F}_n = \bigcap_{n=1}^{\infty} F_n = F.$$

Therefore F is a closed set. Q.E.D.

Closed Set

Theorem 3. If $\{F_\gamma : \gamma \in \Gamma\}$ is a family of closed sets in $\mathbb{R}^N$, then its intersection $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ is a closed set.

Proof. Define $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ where F_γ is a closed set for $\gamma \in \Gamma$. Note $F \subseteq F_\gamma$, then $F' \subseteq F'_\gamma$. Thus $F' \subseteq \bigcap_{\gamma\in\Gamma} F'_\gamma$. Note that F_γ is a closed set, $F'_\gamma \subseteq F_\gamma$, then

$$\left(\bigcap_{n=1}^{\infty} F_n\right)' = F' \subseteq \bigcap_{\gamma\in\Gamma} F'_\gamma \subseteq \bigcap_{\gamma\in\Gamma} F_\gamma = F.$$

Therefore F is a closed set. Q.E.D.

Closed Set

Theorem 3. If $\{F_\gamma : \gamma \in \Gamma\}$ is a family of closed sets in $\mathbb{R}^N$, then its intersection $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ is a closed set.

Proof. Define $F = \bigcap_{\gamma\in\Gamma} F_\gamma$ where F_γ is a closed set for $\gamma \in \Gamma$. Note $F \subseteq F_\gamma$, then $\overline{F} \subseteq \overline{F}_\gamma$. Thus $\overline{F} \subseteq \bigcap_{\gamma\in\Gamma} \overline{F}_\gamma$. Note that F_γ is a closed set, $\overline{F}_\gamma \subseteq \overline{F}$, then

$$\overline{\bigcap_{n=1}^{\infty} F_n} = \overline{F} \subseteq \bigcap_{\gamma\in\Gamma} \overline{F}_\gamma = \bigcap_{\gamma\in\Gamma} F_\gamma = F.$$

Therefore F is a closed set. Q.E.D.

Closed Set

Note that the union of an infinite number of closed sets is not necessarily a closed set. For example, let

$$F_k = \left[\frac{1}{k+1}, \frac{1}{k}\right] \subseteq \mathbb{R} \qquad (k = 1, 2, \cdots),$$

then $\bigcup_{k=1}^{\infty} F_k = (0, 1]$. Furthermore, $\overline{\bigcup_{k=1}^{\infty} F_k} \neq \bigcup_{k=1}^{\infty} \overline{F}_k$.

But we have the following simple facts: Let $E_\gamma \subseteq \mathbb{R}^N$ for $\gamma \in \Gamma$, then

$$\bigcup_{\gamma\in\Gamma} \overline{E}_\gamma \subseteq \overline{\bigcup_{\gamma\in\Gamma} E_\gamma} \qquad \text{and} \qquad \overline{\bigcap_{\gamma\in\Gamma} E_\gamma} \subseteq \bigcap_{\gamma\in\Gamma} \overline{E}_\gamma.$$

Closed Set

Theorem 4. The closure $\overline{A}$ of A is the intersection of all the closed sets which contain A.

Proof. Let E be the intersection of all the closed sets which contain A; by the preceding theorem, E is closed. Since $\overline{A}$ is a closed set which contains A, we have $E \subseteq \overline{A}$. But we also have $E = \overline{E} \supseteq \overline{A}$, by condition (2), whence $E = \overline{A}$. Q.E.D.

Closed Set

Theorem 5 (Cantor Nested Closed Sets Theorem) If $\{F_k; k \in \mathbb{N}\}$ is a sequence of non-empty bounded closed sets in $\mathbb{R}^N$ with $F_1 \supseteq F_2 \supseteq \cdots \supseteq F_k \supseteq \cdots$, then

$$\bigcap_{k=1}^{\infty} F_k \neq \varnothing.$$

Proof. If there are infinitely many identical sets in $\{F_k; k \in \mathbb{N}\}$, then there is a natural number $K \in \mathbb{N}$, define $F_k = F_K$ for $k \geqslant K$.

$$\bigcap_{k=1}^{\infty} = F_K \neq \varnothing.$$

Now let's assume that for all $k \in \mathbb{N}$, $F_{K+1} \subseteq F_k$, and F_{K+1} is the proper subset of F_k, i.e. $F_k \setminus F_{K+1} \neq \varnothing$ for $k \in \mathbb{N}$. We choose $x_k \in F_k \setminus F_{K+1}$ for $k \in \mathbb{N}$, then $\{x_k; k \in \mathbb{N}\}$ is a sequence of bounded distinct points in $\mathbb{R}^N$. According to Bolzano-Weierstrass theorem, there are $\{x_{k_q}; q \in \mathbb{N}\}$ and $x \in \mathbb{R}^N$ such that

$$\lim_{q \to \infty} \left| x_{k_q} - x \right| = 0.$$

Since each F_k is a closed set, then $x \in F_k$ for $k \in \mathbb{N}$, i.e., $x \in \bigcap_{k=1}^{\infty} F_k$.

Q.E.D.

Closed Set

Questions. Answer the following questions:

1. Let $E \subseteq \mathbb{R}$ be a non-empty set. If any subset of E is a closed set, is E a finite set?
2. Let E_1 and E_2 be the sets of $\mathbb{R}$, whether must the equation $\overline{E_1 \cap E_2} = \overline{E}_1 \cap \overline{E}_2$ hold?
3. Let $E_k \subseteq \mathbb{R}^N$ for $k \in \mathbb{N}$, define $E = \bigcup_{k=1}^{\infty} E_k$. If $x_0 \in E'$, whether must there be E_{k_0} such that $x_0 \in E'_{k_0}$?
4. Let $E \subseteq \mathbb{R}^N$. Prove that $\overline{E}$ is the intersection of all closed sets F containing E: $\overline{E} = \bigcap_{F \supseteq E} F$.
5. Let $F \subseteq \mathbb{R}$ be a bounded closed set and $f(x)$ be a (real-valued) function defined on F. If for any $x_0 \in F'$, $f(x) \to +\infty$ as $x \to x_0$ and $x \in F$, prove that F is a countable set.
6. Let $f \in C(\mathbb{R})$, prove that $F = \{(x, y), f(x) \geqslant y\}$ is a closed set in $\mathbb{R}^2$.
7. Try to list countable collection of disjoint dense countable sets in $\mathbb{R}$.

Open Set

Region is a special point set in mathematical analysis. An important proposition for region D is that, if point $P \in D$, then there exists a small area including P in D. Thus every point in D is an interior point.

If $E^\circ = E$, every point in E is interior, then E is an open set.

Theorem: If G is a non-empty set in $\mathbb{R}^N$, then G is an open set if and only if, for any point $x \in G$, there is $\delta > 0$ such that $B(x, \delta) \subseteq G$

The unit open interval $(0, 1)$ is an open set in $\mathbb{R}$, but, in $\mathbb{R}$, interval $[0, 1)$ is not an open set, interval $(0, 1]$ is not an open set, closed interval $[0, 1]$ is not an open set. The unit open circle $\{(x_1, x_2); x_1^2 + x_2^2 < 1\}$ is an open set in $\mathbb{R}^2$, but, in $\mathbb{R}^2$, the unit closed circle $\{(x_1, x_2); x_1^2 + x_2^2 \leqslant 1\}$ is not an open set.

The empty set $\varnothing$ and the whole space $\mathbb{R}$ are open.

Open Set

A correspondence $X \to X^\circ$ is called an interior operation if

(1) $X^\circ \subseteq X$,

(2) $X_1 \subseteq X_2 \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = X^\circ$,

(4) $S^\circ = S$.

Open Set

Theorem 1. If $E \subseteq \mathbb{R}^N$, then E° is an open set.

Proof. Let $E \subseteq \mathbb{R}^N$ and consider any $x_0 \in E^\circ$. Then $N_\delta(x_0) \subseteq E$ for some $\delta > 0$. We claim that $N_\delta(x_0) \subseteq E^\circ$, thus proving E° is open.

To establish the claim consider any $x \in N_\delta(x_0)$; we need to show that x is an interior point of E. Suppose $\rho(x, x_0) = r(< \delta)$. From the triangle inequality, $N_{\delta - r}(x) \subseteq N_\delta(x_0)$, and so $N_{\delta - r}(x) \subseteq E$, thus showing x is an interior point. Q.E.D.

Open Set

Theorem 2. The intersection of finite open sets is an open set.

Proof. Define $G = \bigcap_{n=1}^{N} G_n$ where G_n is an open set for $n = 1, \cdots, N$. Note that G_n is an open set, $G_n^\circ = G_n$, then

$$G^\circ = \left(\bigcap_{n=1}^{N} G_n \right)^\circ = \bigcap_{n=1}^{N} G_n^\circ = \bigcap_{n=1}^{N} G_n = G.$$

Therefore G is an open set. Q.E.D.

Open Set

Theorem 2. The intersection of finite open sets is an open set.

Proof. Define $G = \bigcap_{n=1}^{N} G_n$ where G_n is an open set for $n = 1, \cdots, N$. Suppose $x \in G$, then $x \in A_n$ for $n = 1, \cdots, N$, and for each n there exists $\delta_n > 0$ such that $N_{\delta_n}(x) \subseteq G_n$. Let $\delta = \min\{\delta_1, \cdots, \delta_N\}$. Then $\delta > 0$ and $N_\delta(x) \subseteq G$, implying G is open. Q.E.D.

Open Set

Theorem 3. The union of multiple open sets is an open set.

Proof. Define $G = \bigcup_{n=1}^{\infty} G_n$ where G_n is an open set for $n = 1, 2, \cdots$. Note $G_n \subseteq G$, then $G_n^\circ \subseteq G^\circ$. Thus $\bigcup_{n=1}^{\infty} G_n^\circ \subseteq G^\circ$. Note that G_n is an open set, $G_n^\circ = G_n$, then

$$G = \bigcup_{n=1}^{\infty} G_n = \bigcup_{n=1}^{\infty} G_n^\circ \subseteq G^\circ = \left(\bigcup_{n=1}^{\infty} G_n \right)^\circ .$$

Therefore G is an open set. Q.E.D.

Open Set

Theorem 3. The union of multiple open sets is an open set.

Proof. Define $G = \bigcup_{n=1}^{\infty} G_n$ where G_n is an open set for $n = 1, 2, \cdots$. Suppose $x \in G$, then $x \in G_n$ for some n. For some such n choose $\delta_n > 0$ such that $N_{\delta_n}(x) \subseteq G_n$. Then certainly $N_{\delta_n}(x) \subseteq G$, and so G is open. Q.E.D.

Open Set

Theorem 3. If $\{G_\gamma : \gamma \in \Gamma\}$ is a family of open sets in $\mathbb{R}^N$, then its union $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ is an open set.

Proof. Define $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ where G_γ is an open set for $\gamma \in \Gamma$. Note $G_\gamma \subseteq G$, then $G_\gamma^\circ \subseteq G^\circ$. Thus $\bigcup_{\gamma\in\Gamma} G_\gamma^\circ \subseteq G^\circ$. Note that G_γ is an open set, $G_\gamma^\circ = G_\gamma$, then

$$G = \bigcup_{\gamma\in\Gamma} G_\gamma = \bigcup_{\gamma\in\Gamma} G_\gamma^\circ \subseteq G^\circ = \left(\bigcup_{\gamma\in\Gamma} G_\gamma\right)^\circ.$$

Therefore G is an open set. Q.E.D.

Open Set

Theorem 3. If $\{G_\gamma : \gamma \in \Gamma\}$ is a family of open sets in $\mathbb{R}^N$, then its union $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ is an open set.

Proof. Define $G = \bigcup_{\gamma\in\Gamma} G_\gamma$ where G_γ is an open set for $\gamma \in \Gamma$.

Suppose $x \in G$, then $x \in G_\gamma$ for some $\gamma \in \Gamma$. For some such γ choose $\delta_\gamma > 0$ such that $N_{\delta_\gamma}(x) \subseteq G_\gamma$. Then certainly $N_{\delta_\gamma}(x) \subseteq G$, and so G is open. Q.E.D.

Open Set

The following important theorems reveal the structure of open sets and give us a further perceptual understanding of open sets.

Theorem 4

(1) The non-empty open set in $\mathbb{R}$ is the union of countable disjoint open intervals (here also includes $(-\infty, a)$, (b, ∞) and $(-\infty, \infty)$);

(2) The non-empty open set G in $\mathbb{R}^N$ is the union of countable disjoint semi-open and semi-closed rectangulars.

Open Set

Proof of (1). Let G be an open set in $\mathbb{R}$. For any point $a \in G$, since a is the interior point of G, there is $\delta > 0$ such that $(a-\delta, a+\delta) \subseteq G$. Now define

$$a' = \inf\{x : (x, a) \subseteq G\} \qquad \text{and} \qquad a'' = \sup\{x : (a, x) \subseteq G\},$$

(here, a' can be $-\infty$, and a'' can be $+\infty$). obviously, $a' < a < a''$ and $(a', a'') \subseteq G$. In fact, for any point z in the interval (a', a''), we assume $a' < z \leqslant a$, there must be x such that $a' < x < z$ and $(x, a) \subseteq G$, that is, $z \in G$. We call such an open interval (a', a'') the constituent interval I_a of G (about point a).

If $I_{a_1} = (a_1', a_1'')$, $I_{a_2} = (a_2', a_2'')$ is the constituent interval of G, then they are coincident or disjoint. For this purpose, we assume $a_1 < a_2$. If $I_{a_1} \cap I_{a_2} \neq \varnothing$, then there is $a_2' < a_1''$, then define $\min\{a_1', a_2'\} = a_0$ and $\max\{a_1'', a_2''\} = a^0$, then $(a_0, a^0) = (a_1', a_1'') \cup (a_2', a_2'')$. Take $x \in I_{a_1} \cap I_{a_2}$, then $I_x = (a_0, a^0)$ is a constituent interval and $(a_0, a^0) = (a_1', a_1'') = (a_2', a_2'')$.

Finally, we know that the family of disjoint intervals in $\mathbb{R}$ are countable.

Open Set

Proof of (2). Firstly, $\mathbb{R}^N$ is divided into countable semi-open and semi-closed cubes with side length of 1 by lattice points (coordinates are all integers), and the whole set is denoted as Γ_0. Each side of each cube in Γ_0 is bisected, then each cube can be divided into 2^N semi-open and semi-closed cubes with a side length of $\frac{1}{2}$, In Γ_0, all the sub-cubes thus created are denoted Γ_1. Continue to divide by this method to obtain the sequence $\{\Gamma_k; k \in \mathbb{N}\}$, where the side length of each cube in Γ_k is 2^{-k}, and this cube is the union of the corresponding 2^k disjoint cubes in Γ_{k+1}. We call the cube thus divided a dyadic cube.

Open Set

Now put all cubes in Γ_0 included in G are taken out and written as $\mathscr{H}_0$. Then put all cubes in Γ_1 included in $G \setminus \bigcup_{H\in\mathscr{H}_0} H$ are taken out and written as $\mathscr{H}_1$ ($H \in \mathscr{H}_0$ is semi-open and semi-closed cube). And so on, $\mathscr{H}_K$ is the whole set of semi-open and semi-closed cubes in Γ_K included in $G \setminus \bigcup_{k=1}^{K-1} \bigcup_{H\in\mathscr{H}_k} H$. The whole set of semi-open and semi-closed cubes in $\mathscr{H}_K$ is countable for $K = 0, 1, 2, \cdots$. Since G is an open set, for any $x \in G$, there is $\delta > 0$ such that $B(x, \delta) \subseteq G$. The diameter of the cube in Γ_K approaches to zero as $K \to \infty$, so x will eventually fall into a certain cube in Γ_K. Thus

$$G = \bigcup_{k=0}^{\infty} \bigcup_{H\in\mathscr{H}_k} H, \quad H \text{ denotes semi-open and semi-closed dyadic cubes.}$$

Open Set

There exists another important fact about the open sets in $\mathbb{R}^N$, that is, there is an open set family Γ, which consists of countable open sets in $\mathbb{R}^N$ such that any open set in $\mathbb{R}^N$ is the union of some open sets in Γ. In fact, Γ can be taken as

$$\left\{ B\left(x, \frac{1}{K}\right) \mid x \text{ is a rational point in } \mathbb{R}^N, \text{ and } K \text{ is a natural number} \right\}$$

Firstly, Γ is a countable set. Secondly, for any point x in the open set G in $\mathbb{R}^N$, there must be $\delta > 0$ such that $B(x, \delta) \subseteq G$. Now take the rational point x' such that $d(x, x') < \frac{1}{K}$, where $K > \frac{2}{\delta}$ such that

$$x \in B\left(x', \frac{1}{K}\right) \subseteq B(x, \delta) \subseteq G.$$

Obviously, the union of such $B\left(x', \frac{1}{K}\right)$ is G. Q.E.D.

Open Set

Questions. Answer the following questions:

1. If $N \geqslant 2$, $F \subset \mathbb{R}^N$ $(F \neq \mathbb{R}^N)$ is a closed set, then ∂F is an infinite set.
2. Let $E \subseteq \mathbb{R}^N$, prove that $E^\circ = \mathbf{C}\left(\overline{\mathbf{C}E}\right)$ and $\partial E = \overline{E} \setminus E^\circ$.
3. Prove that the set of discontinuous points of function $f(x_1, x_2) = \begin{cases} x_1 \sin \frac{1}{x_2}, & x_2 \neq 0, \\ 0, & x_2 = 0 \end{cases}$ is not a closed set.
4. Prove that $G \subseteq \mathbb{R}^N$ is an open set if and only if $G \cap \partial G = \varnothing$; $F \subseteq \mathbb{R}^N$ is a closed set if and only if $\partial F \subseteq F$.
5. Let $G \subseteq \mathbb{R}^N$ be a non-empty open set, $r_0 > 0$. If for any $x \in G$, define a closed sphere $\overline{B(x, r_0)}$, Prove that $A = \bigcup_{x \in G} \overline{B(x, r_0)}$ is an open set.
6. Let $F \subseteq \mathbb{R}$ be a closed set, prove that there is a countable subset of E in F such that $\overline{E} = F$.
7. Let every point in $E \subseteq \mathbb{R}^N$ be an isolated point of E, prove that E is the intersection of an open set and a closed set.
8. Let $f \in C([a, b])$, and $f'(x) > 0$ except $\{x_n\} \subseteq [a, b]$, prove that $f(x)$ is strictly increasing.

Open Set

Let $E \subseteq \mathbb{R}^N$, Γ is an family of open sets in $\mathbb{R}^N$. If for any $x \in E$, there exists $G \in \Gamma$ such that $x \in G$, then Γ is called an open covering of E. Let Γ be an open covering of E. If $\Gamma' \subseteq \Gamma$ is still an open covering of E, then Γ' is a subcovering of Γ (about E).

Lemma Any open cover Γ of the set $E \subseteq \mathbb{R}^N$ contains a countable subcovering.

Open Set

Theorem 5 (Heine-Borel Finite Covering Theorem) An open covering of a bounded closed set in $\mathbb{R}^N$ contains a finite subcovering.

Proof Let F is a bounded closed set in $\mathbb{R}^N$, and Γ is an open covering of F. From the above Lemma, it can be assumed that Γ consists of countable open sets:

$$\Gamma = \{G_1, \cdots, G_K, \cdots\}.$$

Define

$$H_K = \bigcup_{k=1}^{K} G_k, \qquad L_K = F \cap \mathbf{C}H_K \quad k = 1, 2, \cdots.$$

Obviously, H_K is an open set, L_K is a closed set with $L_K \supseteq L_{K+1}$ for $k = 1, 2, \cdots$. There are two cases:

(1) there is K_0 such that L_{K_0} is an empty set, that is, $\mathbf{C}H_{K_0}$ does not contain any point of F, then $F \subseteq H_{K_0}$, the theorem is proved;

(2) If all L_K is not an empty set, then, from Cantor's closed set theorem, there is a point of $x_0 \in L_K$ for $k = 1, 2, \cdots$, i.e. $x_0 \in F$ and $x_0 \in \mathbf{C}H_K$ for $k = 1, 2, \cdots$. Thus there is a point $x_0 \in F$ does not belong to everything H_K, which is inconsistent with the original assumption, so case (2) does not exist. Q.E.D.

Open Set

Theorem 5 (Heine-Borel Finite Covering Theorem) If $\mathscr{H}$ is an open covering of a closed and bounded subset E of $\mathbb{R}^N$ (for any x in E there exists a neighborhood $H \in \mathscr{H}$ such that $x \in H$); then E has an open covering $\tilde{\mathscr{H}}$ consisting of finitely many open sets belonging to $\mathscr{H}$.

Proof. The proof method belongs to Lebesgue. We complete the proof by two steps.

Open Set

(1) There exists $\delta > 0$ such that any δ neighborhood $N(x, \delta)$ of center of x in E is contained in some open neighborhood in $\mathscr{H}$. We use reduction to absurdity to check this setting, and assume that there isn't such $\delta > 0$. For any natural number n, we cannot take $\frac{1}{n}$ as δ, then there exists $x_n \in E$ such that $N\left(x_n, \frac{1}{n}\right)$ does not contain in any open neighborhood in $\mathscr{H}$. E is bounded, $\{x_n : n = 1, 2, \cdots\} \subseteq E$ is also bounded. If $\{x_n : n = 1, 2, \cdots\}$ has infinite different members, then there exists an accumulation points x_0 from Bolzano - Weierstrass Theorem. If $\{x_n : n = 1, 2, \cdots\}$ has only finite different members, then there exists at least a point x_0 presented infinite times. In a word, $\{x_n : n = 1, 2, \cdots\} \subseteq E$ has a convergent subsequence $\{x_{n_k} : n = 1, 2, \cdots\} \subseteq E$ with $\lim_{k\to\infty} x_{n_k} = x_0$. Note that E is a closed set, then $x_0 \in E$. Since $\mathscr{H}$ is an open covering of a closed and bounded set E, then there exists $N \in \mathscr{H}$ such that $x_0 \in N$. Define $N = N(y_0, \eta)$, then there exists $\eta' = \min\{\eta, \rho(x_0, y_0)\} > 0$ such that $N(x_0, \eta') \subseteq N(y_0, \eta)$. Note $\lim_{k\to\infty} x_{n_k} = x_0$, then we can take enough large natural number n_k such that $\rho(x_{n_k}, x_0) < \frac{\eta'}{2}$ and $\frac{1}{n_k} < \frac{\eta'}{2}$, then $N\left(x_{n_k}, \frac{1}{n_k}\right) \subseteq N(x_0, \eta') \subseteq N(y_0, \eta) \in \mathscr{H}$. This contradicts the definition of x_{n_k}. Therefore there exists such $\delta > 0$, which is called Lebesgue number.

Open Set

(2) Since E is bounded, we can divide E into finite small cubes by using hyperplanes parallel axises such that the length in the each cube is strictly less than $\delta > 0$. Set these small cubes are $E_1, \cdots, E_K$ with $E = \sum_{k=1}^{K} E_k$, we take arbitrary $x_k \in F_k$ such that $F_k \subseteq N(x_k, \delta)$, then there exists $N_k \in \mathscr{H}$ such that $N(x_k, \delta) \subseteq N_k$. Thus we obtain finite open neighborhoods $N_1, \cdots, N_K$ with

$$E = \sum_{k=1}^{K} E_k \subseteq \bigcup_{k=1}^{K} N(x_k, \delta) \subseteq \bigcup_{k=1}^{K} N_k. \qquad \text{Q.E.D.}$$

Open Set

Note that in the above theorem, the boundedness condition cannot be absent. For example, in $\mathbb{R}$, there is no finite subcovering for open covering $\left\{\left(n-\frac{1}{2}, n+\frac{1}{2}\right)\right\}$ of natural number set.

Similarly, the conditions of a closed set are also indispensable. For example, in $\mathbb{R}$, for the point set $\left\{1, \frac{1}{2}, \cdots, \frac{1}{n}, \cdots\right\}$, define an open covering

$$\left\{\left(\frac{1}{n}-\frac{1}{2n}, \frac{1}{n}+\frac{1}{2n}\right)=\left(\frac{1}{2n}, \frac{3}{2n}\right); n \in \mathbb{N}\right\},$$

there is no finite subcovering.

Open Set

Example: If F is a bounded closed set in $\mathbb{R}^N$, and G is an open set in $\mathbb{R}^N$ and $F \subseteq G$, then there is $\delta > 0$ such that when $|x| < \delta$,

$$F + \{x\} \equiv \{y + x \mid y \in F\} \subseteq G.$$

Proof. For any $y \in F$, as $y \in G$, there is $\delta_y > 0$ such that $B(y, \delta_y) \subseteq G$. As $\left\{ B\left(y, \frac{\delta_y}{2}\right) \mid y \in F \right\}$ constitutes an open covering of F, according to the finite sub-covering theorem, there is $y_1, \cdots, y_K \in F$ such that

$$F \subseteq \bigcup_{k=1}^{K} B\left(y_k, \frac{\delta_{y_k}}{2}\right).$$

Thus, each $y \in F$ belongs to at least one $B\left(y_k, \frac{\delta_{y_k}}{2}\right)$, and the distance between y and any point z in $\mathbf{C}G$ is

$$|y - z| \geqslant |z - y_k| - |y_k - y| > \delta_{y_k} - \frac{\delta_{y_k}}{2} = \frac{\delta_{y_k}}{2}.$$

Take $\delta = \frac{1}{2} \min\{\delta_{y_1}, \cdots, \delta_{y_K}\}$, then, when $|x| < \delta$, $y + x \in G$, that is, $F + \{x\} \subseteq G$.

Open Set

Theorem 6. Let $E \subseteq \mathbb{R}^N$. If any open covering of E contains finite sub-coverings, then E is a bounded closed set.

Proof. If $y \in \mathbf{C}E$, then for every $x \in E$, there is $\delta_x > 0$ such that

$$B(x, \delta_x) \cap B(y, \delta_x) = \varnothing.$$

Obviously, $\{B(x, \delta_x) : x \in E\}$ is an open covering of E. Then there are a finite sub-covering, written as

$$B(x_1, \delta_{x_1}), \cdots, B(x_K, \delta_{x_K}),$$

then E is a bounded set. Now define $\delta_0 = \min\{\delta_{x_1}, \cdots, \delta_{x_K}\}$, then $B(y, \delta_0) \cap E = \varnothing$, that is, $y \notin E'$. Thus $E' \subseteq E$. E is a closed set and hence boundedness is obvious. Q.E.D.

Note that if any open covering of E contains a finite sub-covering, we call E a compact set. The above two theorems show that a compact set in $\mathbb{R}^N$ is a bounded closed set.

Open Set

Questions. Answer the following questions:

1. In $\mathbb{R}^N$, $\{G_\gamma; \gamma \in \Gamma\}$ is an open covering of E. Can $\{\overline{G_\gamma}; \gamma \in \Gamma\}$ cover $\overline{E}$?
2. Let $\Gamma = \{[a_\alpha, b_\alpha] : \alpha \in [0,1]\}$, and any two closed intervals in Γ must intersect. Prove that
$$\bigcap_{\alpha \in [0,1]} [a_\alpha, b_\alpha] \neq \varnothing.$$
3. Let $F \subseteq \mathbb{R}$ be a non-empty countable closed set, prove that F must contain isolated points.
4. Let $\{f_n(x)\}$ be a sequence of non-negative decreasing continuous functions on $\mathbb{R}$. If $f_n(x) \to 0$ as $n \to \infty$ on a bounded closed set F, prove $f_n(x)$ uniformly converges to zero on F.

Close set and Open Set

If with each subset X of a set S we associate a subset $\overline{X}$ of S such that

(1) $X \subseteq \overline{X}$,

(2) $X_1 \subseteq X_2 \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \overline{X}$,

(4) $\overline{\varnothing} = \varnothing$,

then we call the correspondence $X \to \overline{X}$ a closure operation.

In a similar manner, if with each subset X of a set S we associate a subset X° of S such that

(1) $X^\circ \subseteq X$,

(2) $X_1 \subseteq X_2 \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = X^\circ$,

(4) $S^\circ = S$.

then we call the correspondence $X \to X^\circ$ a interior operation.

Close set and Open Set

The sets F such that $F = \overline{F}$ are said to be closed and the sets G such that $G = G^\circ$ are said to be open, with respect to the appropriate operations in each case.

Example. If S denotes the real line $\mathbb{R}$ and $\overline{X}$ the smallest segment $[a, b]$ which contains all the points of the set X, then the correspondence $X \to \overline{X}$ is a closure operation.

Close set and Open Set

Theorem 1. A one-one correspondence can be established between closure operations and interior operations by means of the formulae:

$$X^\circ = \mathbf{C}(\overline{\mathbf{C}X}) \qquad \text{and} \qquad \overline{X} = \mathbf{C}(\mathbf{C}X)^\circ$$

Proof. The correspondence $X \to X^\circ = \mathbf{C}(\overline{\mathbf{C}X})$ is clearly an interior operation, for

(1) $X^\circ = \mathbf{C}(\overline{\mathbf{C}X}) \subseteq \mathbf{C}(\mathbf{C}X) = X$,

(2) $X_1 \subseteq X_2 \Longrightarrow \mathbf{C}X_1 \supseteq \mathbf{C}X_2 \Longrightarrow (\overline{\mathbf{C}X_1}) \supseteq (\overline{\mathbf{C}X_2}) \Longrightarrow X_1^\circ \subseteq X_2^\circ$,

(3) $(X^\circ)^\circ = \mathbf{C}(\overline{\mathbf{C}X}) = \mathbf{C}(\overline{\mathbf{C}X}) = X^\circ$,

(4) $S^\circ = \mathbf{C}\overline{\varnothing} = \mathbf{C}\varnothing = S$.

Close set and Open Set

The correspondence $X \to \overline{X} = \mathbf{C}(\mathbf{C}X)^\circ$ is clearly a closure operation, for

(1) $\overline{X} = \mathbf{C}(\mathbf{C}X)^\circ \supseteq \mathbf{C}(\mathbf{C}X) = X$,

(2) $X_1 \subseteq X_2 \Longrightarrow \mathbf{C}X_1 \supseteq \mathbf{C}X_2 \Longrightarrow (\mathbf{C}X_1)^\circ \supseteq (\mathbf{C}X_2)^\circ \Longrightarrow \overline{X}_1 \subseteq \overline{X}_2$,

(3) $\overline{\overline{X}} = \mathbf{C}[(\mathbf{C}X)^\circ]^\circ = \mathbf{C}(\mathbf{C}X)^\circ = \overline{X}$,

(4) $\overline{\varnothing} = \mathbf{C}S^\circ = \mathbf{C}S = \varnothing$.

It is easily shown that the correspondence thus set up between the two kinds of operations is one-one. Q.E.D.

Close set and Open Set

Remark. The complement of an open set is a closed set (and vice versa), for $\mathbf{C}F = \mathbf{C}\overline{F} = (\mathbf{C}F)^\circ$; in what follows, we shall study only properties of closure operations, those of interior operations being deducible immediately on appealing to duality, for $\mathbf{C}G = \mathbf{C}G^\circ = \overline{\mathbf{C}G}$.

Close set and Open Set

Theorem 2. The intersection of a family $\mathscr{F} = (F_\gamma \mid \gamma \in \Gamma)$ of closed sets is a closed set.

Proof. Writing $F = \bigcap_{\gamma \in \Gamma} F_\gamma$, we have, by condition (2) for a closure operation,

$$\overline{F} \subseteq \overline{F}_\gamma = F_\gamma$$

for each γ. Therefore $\overline{F} \subseteq F$; hence, by condition (1), $\overline{F} = F$. Q.E.D.

Close set and Open Set

Theorem 2. The union of a family $\mathscr{G} = (G_\gamma \mid \gamma \in \Gamma)$ of open sets is an open set.

Proof. Writing $G = \bigcup_{\gamma \in \Gamma} G_\gamma$, we have, by condition (2) for a interior operation,

$$G_\gamma = G_\gamma^\circ \subseteq G^\circ$$

for each γ. Therefore $G \subseteq G^\circ$; hence, by condition (1), $G^\circ = G$. Q.E.D.

Close set and Open Set

Theorem 3. The closure $\overline{X}$ of X is the intersection of all the closed sets which contain X.

Proof. Let E be the intersection of all the closed sets which contain X; by the preceding theorem, E is closed. Since $\overline{X}$ is a closed set which contains X, we have $E \subseteq \overline{X}$. But we also have $E = \overline{E} \supseteq \overline{X}$, by condition (2), whence $E = \overline{X}$. Q.E.D.

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; a point $x \in \mathbb{R}^N$ is adherent to X if there is a sequence of points of X tending to x. One can also say, in a looser manner, that x is adherent to X if there are points of X arbitrarily close to x. Any point x of X is adherent to X; it suffices to take the sequence whose points are all equal to x.

Limits in $\mathbb{R}^N$

The set of the points of $\mathbb{R}^N$ adherent to X is called the adherence (often also the closure) of X, and denoted $\overline{X}$. According to the last remark of 1.6.g:

(1) $X \subseteq \overline{X}$.

It is also clear that:

(2) $X \subseteq X'$ implies $\overline{X} \subseteq \overline{X'}$.

Example: (1) Let M be a positive real number, and consider, in $\mathbb{R}^2$, the square (with perimeter excluded) $X = \{x \in \mathbb{R}^2 \mid |x_n| < M \text{ for } n = 1, 2\}$; then $\overline{X} = \{x \in \mathbb{R}^2 \mid |x_n| \leqslant M \text{ for } n = 1, 2\}$. (2) Using (2) of 1.5.I, one obtains $\overline{\mathbb{Q}} = \mathbb{R}$: the adherence of the set of rationals is the set of reals.

Limits in $\mathbb{R}^N$

A subset X of $\mathbb{R}^N$ is closed if it owns its adherent points. This may be expressed as $\overline{X} \subseteq X$. Thus, because of (1), X is closed if and only if it is equal to its adherence, $X = \overline{X}$. One may also say that a set is closed if and only if every point at zero distance from the set belongs to the set; this requirement will often be imposed below.

Example: (1) In the examples of 1.6.h. X, the subset of $\mathbb{R}^2$, and $\mathbb{Q}$, the subset of $\mathbb{R}$, are not dosed; $\overline{X}$ is closed. (2) The subset of $\mathbb{R}$, $\left\{0, 1, \frac{1}{2}, \frac{1}{3}, \cdots, \frac{1}{q}, \cdots\right\}$, is closed.

Limits in $\mathbb{R}^N$

One can prove:

(3) If X is a subset of $\mathbb{R}^N$, its adherence $\overline{X}$ is closed.

Proof: For any $\varepsilon > 0$ and any $x \in \overline{\overline{X}}$, there exists $x_\varepsilon \in \overline{X}$ such that $d(x, x_\varepsilon) < \frac{\varepsilon}{2}$ and then there exists $x'_\varepsilon \in X$ such that $d(x_\varepsilon, x'_\varepsilon) < \frac{\varepsilon}{2}$, hence $d(x, x'_\varepsilon) \leqslant d(x, x_\varepsilon) + d(x_\varepsilon, x'_\varepsilon) < \varepsilon$, which implies $x \in \overline{X}$. Hence $\overline{\overline{X}} \subseteq \overline{X}$. $\overline{X}$ is closed. $\square$

That is, $\overline{X} = \overline{\overline{X}}$. Since every closed set containing X clearly contains $\overline{X}$, the adherence of X can be characterized as the smallest closed set containing X.

Limits in $\mathbb{R}^N$

It is easy to prove that:

(4) If $\mathscr{X}$ is a set of closed subsets of $\mathbb{R}^N$, their intersection, $\bigcap_{X\in\mathscr{X}} X$, is a closed subset of $\mathbb{R}^N$.

Proof: For any $x \in \overline{\bigcap_{X\in\mathscr{X}} X}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq \bigcap_{X\in\mathscr{X}} X$ such that $x^q \to x$. And for any $X \in \mathscr{X}$, $\{x^q; q \in \mathbb{N}\} \subseteq X$, which implies $x \in \overline{X}$ for any $X \in \mathscr{X}$. As X is closed, $x \in X$ for any $X \in \mathscr{X}$. Then $x \in \bigcap_{X\in\mathscr{X}} X$, which implies $\overline{\bigcap_{X\in\mathscr{X}} X} \subseteq \bigcap_{X\in\mathscr{X}} X$. So $\bigcap_{X\in\mathscr{X}} X$ is a closed subset of $\mathbb{R}^N$. $\square$

Limits in $\mathbb{R}^N$

It is a little more difficult to prove that:

(5) If $X_1, \cdots, X_n, \cdots, X_N$ ore N closed subsets of $\mathbb{R}^N$, their union, $\bigcup_{n=1}^{N} X_n$, is a closed subset of $\mathbb{R}^N$.

Proof: For any $x \in \overline{\bigcup_{n=1}^{N} X_n}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq \bigcup_{n=1}^{N} X_n$ such that $x^q \to x$. So there exists $n_0 \in \{1, \cdots, N\}$ such that there are infinite elements $\{x^{q_k}; q_k \in \mathbb{N}\}$ of $\{x^q; q \in \mathbb{N}\}$ in X_{n_0}, hence $x^{q_k} \to x$, which implies $x \in \overline{X}_{n_0}$. As X_{n_0} is closed, $x \in X_{n_0}$. Hence $x \in \bigcup_{n=1}^{N} X_n$. $\bigcup_{n=1}^{N} X_n$ is a closed subset of $\mathbb{R}^N$. $\square$

Limits in $\mathbb{R}^N$

Let S be a subset of $\mathbb{R}^N$; for a subset X of S, one defines the adherence of X in S as the set of points of S adherent to X. Similarly X is said to be closed in S if it owns the points of S adherent to X.

Example: (1) Let M be a positive real number, and take $S = \{x \in \mathbb{R}^2 \mid |x_n| < M \text{ for } n = 1, 2\}$, $X = \{x \in S \mid x_1 \geqslant 0\}$. From a sketch it will be clear that the set X is closed in S but not closed in $\mathbb{R}^2$. (2) Note also that any subset T of $\mathbb{R}^N$ is closed in T.

Limits in $\mathbb{R}^N$

Substituting "the adherence in S" for "the adherence," "closed in S" for "closed," and S for $\mathbb{R}^N$ everywhere in (1) to (5), one obtains new theorems with the same proofs.

Limits in $\mathbb{R}^N$

One can prove:

(6) Every subset S of $\mathbb{R}^N$ contains a countable subset X such that $S \subseteq \overline{X}$.

Proof: As $\overline{\mathbb{Q}^N} = \mathbb{R}^N$ and $\mathbb{Q}^N$ is countable, let $\mathbb{Q}^N = \left\{q^1, q^2, \cdots\right\}$ and

$$T = \left\{(n,k) \in \mathbb{N}^2 : S \cap B\left(q^n, \frac{1}{k}\right) \neq \varnothing\right\}.$$

For every $(n,k) \in T$, let

$$y_{n,k} \in S \cap B\left(q^n, \frac{1}{k}\right),$$

then

$$X = \{y_{n,k} : (n,k) \in T\}$$

is a countable subset of S. Then we need to show $S \subseteq \overline{X}$.

Limits in $\mathbb{R}^N$

Suppose any $x \in S$ and any $\varepsilon > 0$, there exists $k \in \mathbb{N}$ such that $\frac{2}{k} < \varepsilon$. Since $\overline{\mathbb{Q}^N} = \mathbb{R}^N$, there exists n so that

$$q^n \in B\left(x, \frac{1}{k}\right)$$

This implies $x \in B\left(q^n, \frac{1}{k}\right)$. So $(n, k) \in T$ and $y_{n,k} \in X$ and

$$d\left(x, y_{n,k}\right) < \frac{2}{k} < \varepsilon.$$

Hence $S \subseteq \overline{X}$. $\square$

Limits in $\mathbb{R}^N$

In other words, an arbitrary (therefore perhaps non-countable) subset S of $\mathbb{R}^N$ contains a countable set X which is dense in S, i.e., such that for any point x of S there are points of X arbitrarily close to x.

Example: Let S be $\mathbb{R}^N$ itself. From $\overline{\mathbb{Q}} = \mathbb{R}$ (example (2) of 1.6.h) follows $\overline{\mathbb{Q}^N} = \mathbb{R}^N$. Thus $\mathbb{Q}^N$, the product of N sets equal to $\mathbb{Q}$, i.e. the sets of points of $\mathbb{R}^N$ whose coordinates are all rationals, is dense in $\mathbb{R}^N$. Moreover, $\mathbb{Q}^N$ is countable. (apply (2) of 1.3.n to the N countable sets $\mathbb{Q}$).

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; a point $x \in \mathbb{R}^N$ is interior to X if it is not adherent to the complement of X. This means, intuitively, that x is completely surrounded by points of X. Any point x of $\mathbb{R}^N$ interior to X clearly belongs to X. The interior of X is the set of points interior to X, viz., $\mathbf{C}\left(\overline{\mathbf{C}X}\right)$.

Example: (1) Let M be a positive real numbers, and consider, in $\mathbb{R}^2$, the closed square $X = \left\{x \in \mathbb{R}^2 \mid |x_n| \leqslant M \text{ for } n = 1, 2\right\}$; its interior is $\left\{x \in \mathbb{R}^2 \mid |x_n| < M \text{ for } n = 1, 2\right\}$. (2) The set $\mathbb{Q}^2$ and the straight line $Y = \left\{y \in \mathbb{R}^2 \mid y_2 = 0\right\}$ in $\mathbb{R}^2$ have empty interiors.

Limits in $\mathbb{R}^N$

As in the case of 1.6.I, if X is a subset of S (a subset of $\mathbb{R}^N$), a point $x \in S$ is said to be interior to X in S if it is not adherent to $\mathbf{C}_S X$, the complement of X in S.

Example: (3) Consider, in $\mathbb{R}^2$, the straight line $S = \{z \in \mathbb{R}^2 \mid z_1 = 0\}$ and its subset $Z = \{z \in \mathbb{R}^2 \mid z_1 = 0, 0 \leqslant z_2 \leqslant 1\}$. The interior of Z in S is $\{z \in \mathbb{R}^2 \mid z_1 = 0, 0 < z_2 < 1\}$; its interior in $\mathbb{R}^2$ is empty.

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; a point in $\mathbb{R}^N$ is a boundary point of X if it is adherent to both X and its complement $\mathbf{C}X$. The boundary of X is the set of its boundary points, viz., $\overline{X} \cap \overline{\mathbf{C}X}$.

Example: In the last group of examples, the boundary of the closed square X is its perimeter, that of $\mathbb{Q}^2$ is $\mathbb{R}^2$, that of Y is Y.

Limits in $\mathbb{R}^N$

Let X be a subset of $\mathbb{R}^N$; the exterior of X is the complement of its adherence, via., $\mathbf{C}\overline{X}$. It is easy to prove that the interior, the boundary, and the exterior of X form a partition of $\mathbb{R}^N$ or, what is equivalent, that the interior and the boundary of X form a partition of the adherence of X. The last remark shows that the set X is closed if and only if it contains its boundary.

Limits in $\mathbb{R}^N$

A subset S of $\mathbb{R}^N$ is said to be bounded if it is contained in some closed cube $\mathbb{M}$. A set which is unbounded is therefore a set which has points arbitrarily far from the origin.

Example: The set $\mathbb{J}$ of integers (a subset of $\mathbb{R}$) is unbunded.

Limits in $\mathbb{R}^N$

A subset S of $\mathbb{R}^N$ is said to be compact if it is closed and bounded. It is easily proved that:

(7) If $S_1, \cdots, S_k, \cdots, S_K$ are closed (resp. compact) subsets of $\mathbb{R}^{N_1}, \cdots, \mathbb{R}^{N_k}, \cdots, \mathbb{R}^{N_K}$, then $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is a closed (resp. compact) subset of $\mathbb{R}^{\sum_{k=1}^{K} N_k}$.

Proof: Let $x = (x_1, \cdots, x_k, \cdots, x_K) \in S_1 \times \cdots \times S_k \times \cdots \times S_K$ where $x_k \in S_k$ for $k = 1, \cdots, K$.

For any $x \in \overline{S_1 \times \cdots \times S_k \times \cdots \times S_K}$, there exists $\{x^q; q \in \mathbb{N}\} \subseteq S_1 \times \cdots \times S_k \times \cdots \times S_K$ such that $x^q \to x$. Hence for any $k = 1, \cdots, K$, we have $x_k^q \to x_k$. As S_k is closed, $x_k \in S_k$. Hence $x \in S_1 \times \cdots \times S_k \times \cdots \times S_K$. So $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is a closed.

If $S_1, \cdots, S_k, \cdots, S_K$ are compact, then for $k = 1, \cdots, K$, any $x_k \in S_k$, there exists $M_k \geqslant 0$ such that $\|x_k\| \leqslant M_k$. Let $M = \max\{M_1, \cdots, M_k, \cdots, M_K\}$, then for any $x = (x_1, \cdots, x_k, \cdots, x_K) \in S_1 \times \cdots \times S_k \times \cdots \times S_K$, we have $\|x\| \leqslant M$, i.e. $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is bounded. As $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is also closed, we have that $S_1 \times \cdots \times S_k \times \cdots \times S_K$ is compact. $\square$

Limits in $\mathbb{R}^N$

Finally, a subset S of $\mathbb{R}^N$ is said to be connected if it is not the union of two non-empty, disjoint subsets closed in S. In other words, a set S is connected if it cannot be partitioned into two non-empty subsets closed in S, i.e., loosely speaking, if it is of one piece (but perhaps with holes). One can prove that:

(8) A subset of $\mathbb{R}$ is connected if and only if it is an interval.

Limits in $\mathbb{R}^N$

Proof: Let A is connected. Assume A is not an interval, then we can take three real numbers a_1, a_2, a such that $a_1 < a < a_2$, $a_1 \in A$, $a_2 \in A$ and $a \notin A$. Let $A_1 = A \cap (-\infty, a]$, $A_2 = A \cap [a, \infty)$, then $A = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$. Notice that $(-\infty, a]$ and $[a, \infty)$ are both non-empty closed subsets of $\mathbb{R}$, hence A_1 and A_2 are closed in A, hence A is therefore not connected. Contradiction!

Conversely let A be an interval of $\mathbb{R}$. To derive a contradiction, suppose we could write $A = A_1 \cup A_2$ for nonempty and disjoint closed subsets A_1 and A_2 of A. Pick any $a_1 \in A_1$ and $a_2 \in A_2$ and let $a_1 < a_2$ without loss of generality. Define $a = \sup\{t \in A_1 : t < a_2\}$ [or $a = \inf\{t \in A_2 : a_1 < t\}$], and note that $a_1 \leqslant a \leqslant a_2$ and hence $a \in A$ (because A is an interval).

If $a \in A_1$, then $a \neq a_2$ (since $A_1 \cap A_2 = \varnothing$), so $a < a_2$. As $a \notin A_2$ and A_2 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_2$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a, a + \varepsilon)$, $a_1 < x < a_2$, hence $x \notin A_1 \cup A_2$, i.e. $x \notin A$. Contradiction!

If $a \in A_2$, then $a \neq a_1$ (since $A_1 \cap A_2 = \varnothing$), so $a_1 < a$. As $a \notin A_1$ and A_1 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_1$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a - \varepsilon, a)$, $a_1 < x < a_2$, hence $x \notin A_1 \cup A_2$, i.e. $x \notin A$. Contradiction! $\square$

Connectedness

The two sets A_1 and A_2 in metric space X are said to be separated if

$$(A_1 \cap \overline{A}_2) \cup (A_2 \cap \overline{A}_1) = \varnothing$$

If there exist two non-empty separated sets A_1 and A_2 in metric space X such that $X = A_1 \cup A_2$, then X is a disconnected space. Otherwise, X is a connected space, i.e., there are not two non-empty separated sets A_1 and A_2 in metric space X such that $X = A_1 \cup A_2$.

Specifically, a subset S of metric space X is said to be connected if it is not the union of two non-empty separated sets in S. In other words, a set S is connected if it cannot be partitioned into two non-empty separated sets in S, i.e., loosely speaking, if it is of one piece (but perhaps with holes).

Connectedness

(8) A subset A of $\mathbb{R}$ is connected if and only if it is an interval.

Proof. Let A is connected. Assume A is not an interval, then we can take three real numbers a_1, a_2, a such that $a_1 < a < a_2$, $a_1 \in A$, $a_2 \in A$ and $a \notin A$. Let $A_1 = A \cap (-\infty, a]$, $A_2 = A \cap [a, \infty)$, then $A = A_1 \cup A_2$ and $A_1 \cap A_2 = \varnothing$. Notice that $(-\infty, a]$ and $[a, \infty)$ are both non-empty closed subsets of $\mathbb{R}$, hence A_1 and A_2 are closed in A, hence A is therefore not connected. Contradiction!

Conversely let A be an interval of $\mathbb{R}$. To derive a contradiction, suppose we could write $A = A_1 \cup A_2$ for nonempty and disjoint closed subsets A_1 and A_2 of A. Pick any $a_1 \in A_1$ and $a_2 \in A_2$ and let $a_1 < a_2$ without loss of generality. Define $a = \sup\{t \in A_1 : t < a_2\}$, and note that $a_1 \leqslant a \leqslant a_2$ and hence $a \in A$ (because A is an interval).

If $a \in A_1$, then $a \neq a_2$ (since $A_1 \cap A_2 = \varnothing$), so $a < a_2$. As $a \notin A_2$ and A_2 is closed, then we know there exists $\varepsilon > 0$ such that for all $x \in A_2$, we have $|x - a| > \varepsilon$. Hence for all $x \in (a, a + \varepsilon)$, $a_1 < a < a_2$, hence $x \notin A_1 \cap A_2$, i.e. $x \notin A$. Contradiction!

If $a \in A_2$, then $a > a_1$. Using similar method, we can also get to the contradiction! $\square$

Connectedness

Theorem: For a metric space X, the following statements are equivalent:

(1) X cannot be divided into two disjoint non-empty open sets.

(2) X cannot be divided into two disjoint non-empty closed sets. [That is, X is connected.]

(3) The only clopen subsets of X are X and the empty set. (A clopen set is both open and closed)

(4) The only subsets of X with empty boundary are X and the empty set.

(5) X cannot be written as the union of two non-empty separated sets (sets for which each is disjoint from the other's closure).

(6) All continuous functions from X to $\{0, 1\}$ are constant, where $\{0, 1\}$ is the two-point space endowed with the discrete metric.

Connectedness

Proof of (1) $\Longrightarrow$ (2): If (2) is false, then there exists non-empty, disjoint and closed sets A_1 and A_2 such that $X = A_1 \cup A_2$. Then $\varnothing = A_1 \cap A_2$, hence $X = \mathbf{C}\varnothing = \mathbf{C}A_1 \cup \mathbf{C}A_2$ where $\mathbf{C}A_1$ and $\mathbf{C}A_2$ are disjoint, non-empty and open sets. Thus (1) is false.

Proof of (2) $\Longrightarrow$ (1): If (1) is false, then there exists non-empty, disjoint and open sets A_1 and A_2 such that $X = A_1 \cup A_2$. Then $\varnothing = A_1 \cap A_2$, hence $X = \mathbf{C}\varnothing = \mathbf{C}A_1 \cup \mathbf{C}A_2$ where $\mathbf{C}A_1$ and $\mathbf{C}A_2$ are disjoint, non-empty and closed sets. Thus (2) is false.

Proof of (1) $\Longrightarrow$ (3) and (2) $\Longrightarrow$ (3): If the clopen subset S of X is a proper subset ($S \notin \{\varnothing, X\}$), then $X = S \cup \mathbf{C}S$, where $\mathbf{C}S$ is a clopen proper subset of X, thus X is not connected.

Proof of (3) $\Longrightarrow$ (1): If (1) is false, then there exist non-empty, disjoint and open sets A_1 and A_2 such that $X = A_1 \cup A_2$, hence $A_2 = \mathbf{C}A_1$ is closed. Thus A_2 a clopen proper subset of X, that is, (3) is false.

Proof of (3) $\Longrightarrow$ (2): If (2) is false, then there exist non-empty, disjoint and closed sets A_1 and A_2 such that $X = A_1 \cup A_2$, hence $A_2 = \mathbf{C}A_1$ is open. Thus A_2 a clopen proper subset of X, that is, (3) is false.

Connectedness

Proof of (3) $\Longleftrightarrow$ (4): $A \subseteq X$ with $\partial A = \varnothing$ if and only if A is a clopen subset of X.

$A \subseteq X$ with $\partial A = \varnothing$ if and only if $\overline{A} = \overset{\circ}{A} \cup \partial(A)$ with $\partial A = \varnothing$ if and only if $\overline{A} = \overset{\circ}{A}$ if and only if $\overline{A} = A = \overset{\circ}{A}$ if and only if A is a clopen subset of X.

Proof of (2) $\Longrightarrow$ (5): If (5) is false, then there exist non-empty separated sets A_1 and A_2 such that $X = A_1 \cup A_2$. Since $\overline{A_1} \cap A_2 = \varnothing$ and $\overline{A_2} \cap A_1 = \varnothing$, then $\overline{A_1} \subseteq A_1$ and $\overline{A_2} \subseteq A_2$, i.e. A_1 and A_2 are non-empty closed. Thus (2) is false.

Proof of (5) $\Longrightarrow$ (3): If (3) is false, then there exist a non-empty clopen proper subset A of X, hence $X = A \cup \mathbf{C}A$ where $\mathbf{C}A$ is a non-empty clopen proper subset of X. Thus A and $\mathbf{C}A$ are non-empty separated sets, that is, (5) is false.

Proof of (3) $\Longrightarrow$ (6): As $\{0\}$ and $\{1\}$ are clopen proper subsets in the space $\{0, 1\}$, two-point space endowed with the discrete metric. Then if f is continuous, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen subsets, then $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are either X or $\varnothing$, which implies f is constant.

Proof of (6) $\Longrightarrow$ (3): As $\{0\}$ and $\{1\}$ are clopen proper subsets in the space $\{0, 1\}$, two-point space endowed with the discrete metric. Then if f is continuous, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are clopen subsets. Then if f is constant, $f^{-1}(\{0\})$ and $f^{-1}(\{1\})$ are either X or $\varnothing$. Therefore the only clopen subsets of X are X and the empty set. $\square$

Connectedness

